Holonomic D-modules:

$f: X \to Y$

$0 \to \pi_Y^* \to \pi_X^* \to \pi_{X/Y}^* \to 0$ splits

Thm: (preservation) $f^!, f_\bullet$ preserve holonomicity

Proof: Do cases where f is smooth, f is cl. embed.

① $f^!$ is smooth $\rightsquigarrow$

$$\begin{tikzcd} \pi_Y^* \arrow[d, phantom, "\cup"] & \pi_Y \times_Y X \arrow[l, "\text{smooth}", "\dim(X)-\dim(Y)"'] \arrow[r, hook] & \pi_X \\ SS(M) \arrow[rr, squiggly] & & SS(f^! M) \\ \dim(Y) & \dim(X) & \dim(X) \end{tikzcd}$$

$\boxed{i^! M \text{ hol} \Rightarrow M \text{ is}}$

② i is closed embed, $i^!$

$$i_* i^! M \to M \to j_* j^! M \Rightarrow i_* i^! M \text{ holonomic}$$

lemma; holonomic; holonomic; holonomic

(dimension argument)

Claim: $i_* N$ is holonomic iff N is

holonomic

③ f_* f closed embedding,

④ f_* f smooth + proper.

$$\begin{tikzcd} X \times S \arrow[r, hook, "j"] \arrow[dr, "f"'] & X \times \bar{S} \arrow[d] \\ & X \end{tikzcd}$$

$k \cap U$; $j_* M$ proper

$p^! M$ holonomic

$$\begin{tikzcd} \bar{S} \arrow[r, "p"] \arrow[d, "q"'] & X \times \bar{S} \arrow[d] & M \arrow[d, squiggly] \\ x \arrow[r] & X & f_* M \end{tikzcd}$$

$i_x^! f_* M$ needs to be finite.

i_x^{-1} stalk'l: colim $\mathcal{F}(U)$, $U \ni x$ Zariski topology

$i_x: \{x\} \hookrightarrow X$ $\quad i^! \mathcal{F} \leftarrow i^{\circ}$ - i° pullback.

Ex: $M = \mathcal{O}_X$ $\quad M_x$ big local ring

k

$k[x]$

Claim: If q proper (which it is), M coherent / D_X

$q_* M$ is coherent / D_X.

proper $\longleftrightarrow$ projective

Exercise: • reduce to case X projective.

• Use the relative Spencer resolution, that complex is finite / D_X

But: coherent / D_{pt} $\longleftrightarrow$ finite dim

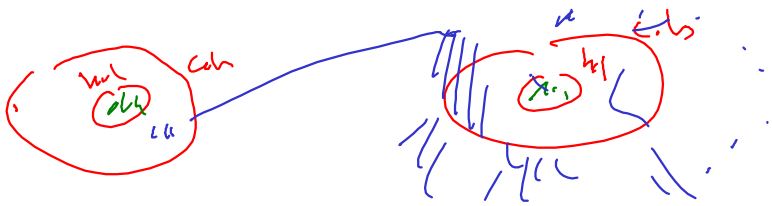
Def: $f: X \to Y$

$f^* = \mathbb{D}_X \circ f^! \circ \mathbb{D}_Y$ $\qquad$ $f_! = \mathbb{D}_Y \circ f_* \circ \mathbb{D}_X$

$D_h DMod(Y) \to D_h DMod(X)$ $\qquad$ $D_h DMod(X) \to D_h DMod(Y)$

- functor on derived cat of holonomic D-mods.
- Definitions seem easy, why so much work? $\mathbb{D}_X, \mathbb{D}_Y$ only well behaved on Dcoherent mod.

But, f_* is not guaranteed to preserve D-coh.
$f^!$



Statements w/o proof:

① f is proper, then $f_* = f_!$

② (f^*, f_*) adjoint
$(f_!, f^!)$ adjoint

Exercise: check it for closed immersion

Exercise: Prove using functor:

$$R\mathcal{H}om_{\mathbb{D}_X}(M, N) \simeq (\omega_X \overset{L}{\otimes} \mathbb{D}_X M)[-\dim X] \overset{L}{\underset{\mathbb{D}_X}{\otimes}} N$$

defn of $\mathbb{D}_X$...

Def^n $f: S \hookrightarrow X$ locally closed emb.

$f_! \longrightarrow f_*$.

- f closed embed, $f_! = f_*$ ... identity
- $f = j$ open: $j^* = j^!$

$M \xrightarrow{\simeq} j^! j_! M$ $\qquad$ $j^* j_* M \xrightarrow{\simeq} M$

$\boxed{f_! \longrightarrow f_*}$

$id \to f^! f_*$ $\qquad$ $f^* f_! \to id$
$\quad \| \qquad\qquad \|$
$f^* f_* \qquad f^! f_!$

inverse of above isos

Exercise: Show they're the same using duality.

Define: middle extension
minimal extension

Assume f is affine $\Rightarrow$ $f_*, f_!$ preserve abelian cat DMod

$$f_{!*}\mathcal{M} = \mathrm{image}(f_!\mathcal{M} \to f_*\mathcal{M})$$

$$= \mathrm{image}(H^0(f_!\mathcal{M}) \to H^0(f_*\mathcal{M}))$$?

$f: S \hookrightarrow X$ affine locally closed.

Thm $\mathcal{E}$ simple flat connection on S

Then,

$L(S,\mathcal{E})$ simple — dual

- $i_{!*}\mathcal{E}$ unique simple quotient of $i_!\mathcal{E}$
- $i_{!*}\mathcal{E}$ unique simple sub of $i_*\mathcal{E}$

} follows from ①

- unique composition factor that has support in S of $i_*\mathcal{E}$ or $i_!\mathcal{E}$

Furthermore, every simple arises this way for some $(S,\mathcal{E})$

Pf

Choose open U of X such that

① $i_!\mathcal{E}$ has unique simple quotient

$U = X -$ Kashiwara

not smooth.

$$S \xrightarrow{i} \bar{S} \xrightarrow{id} X$$

$$\mathcal{E} = i^!(i_!\mathcal{E}) = \mathcal{E}$$ simple.

i.e. proves ④ (restrict to open = exact)

WTS there is unique simple quot.

If $i_!\mathcal{E} \twoheadrightarrow \mathcal{M}$ simple quotient, by exactness.

there is a surjection like this.

$\mathcal{E} = i^!i_!\mathcal{E} \twoheadrightarrow i^!\mathcal{M}$ surjection, so iso ($\mathcal{E}$ is simple)

or $i^!\mathcal{M} = 0$.

$i^!\mathcal{M}$ must be $\mathcal{E}$

$$\boxed{\begin{aligned}&\mathrm{Hom}(i_!\mathcal{E}, \mathcal{M}) \\ &= \mathrm{Hom}(\mathcal{E}, i^!\mathcal{M})\end{aligned}}$$

+ only one component that restricts to $\mathcal{E}$

If $i^!\mathcal{M} = 0$, then $i_!\mathcal{E} \xrightarrow{=0} \mathcal{M}$ iff $\mathcal{M} = 0$.

* <u>Conclusion</u>: There is a unique single quot. of $i_1\mathcal{E}$ wrt .. subs of $i_0\mathcal{E}$

$$i_1\mathcal{E} \longrightarrow i_{1*}\mathcal{E} \longrightarrow i_0\mathcal{E}$$

<u>not zero</u> b/c when restrict, not zero.